

Gradual Deletion, Null Allomorphy, and Variation in Hixkaryana Nasality Agreement

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Abstract

The Hixkaryana language shows mysterious reductions preceding nasal consonants, including optional debuccalization of obstruents and null allomorphy of prefixes. The two processes can be effectively analyzed as minimally distinct extensions of a common response to nasality disagreement. McCarthy's Gradual Deletion framework is leveraged to explain the relationship between these extensions, and ultimately shows us that prefix deletion is best analyzed as part of the same fundamental process as debuccalization. Variation in when and where the process occurs feeds speculation on its evolutionary history, which in turn informs a working synchronic implementation based on the intersection of Gradual Deletion and Output-Output Faithfulness.

1 Introduction

Hixkaryana is a Parukotoan language in the greater Cariban family spoken by around 500-1,000 people in the Nhamundá-Mapuera Indigenous Land in the Amazon rainforest (De Queiroz [2]). While better known for its typologically unique syntax, the language is rich in morphophonology worth investigating. Existing documentation of Hixkaryana's phonological grammar (Derbyshire [3]) is mostly descriptive and does not provide any speculation on the underlying motivations for various processes, including interactions between syncope, nasality agreement, and null allomorphy.

We will see that when obstruents (κ) come in contact with a following nasal (N), one of the two segments is typically forced to change. When κ reduces pre-nasally, it does so to variable extents (resulting in either $?$ or $\emptyset$) depending on the context, creating somewhat of a gradient of possible reductions.

McCarthy [9, 11] observed that speech segments are often shy to delete all of their content at once, resulting in a nested relationship between reduction and deletion. McCarthy's more radical extension of these observations was that full deletion ($\kappa \rightarrow \emptyset$) may actually take place gradually (with reduction as a necessary intermediate step), even in contexts where partial deletion (e.g. reducing to $?$) is never surface-true.

The Gradual Deletion claim is typically motivated by facts of distribution and architectural efficiency - for example, there is an asymmetry in Classical Arabic whereby codas are always the preferred deletion target in the handling of stop clusters. McCarthy gives the example of

/patka/ → [pa.ka]. In a parallel derivation, [pa.ta] and [pa.ka] should be equally optimal outputs, but in reality only [pa.ka] is seen. Serializing the derivation into /patka/ → |paʔ.ka| (→ [pa.ka]) guarantees the correct output by providing an initial evaluation capable of referencing prosody, which will leverage established assumptions of coda weakness, namely that *|pat.ʔa| is inferior because there remains copious segmental content |t| in coda position. The coda is made weak via debuccalization, and has no choice but to collapse into ∅ in the second evaluation.

For vowels across several Arabic varieties in which syncope occurs, there is even more compelling evidence that deletion is taking place gradually because partially and fully deleted segments appear in minimally distinct environments. In these varieties, high vowels will typically syncopate interconsonantly (i → ∅) unless the surrounding consonants match ($C_\alpha i C_\alpha$), which would force illegal gemination. As an alternative solution, high vowels instead reduce to schwa ($C_\alpha i C_\alpha \rightarrow C_\alpha \partial C_\alpha$) (McCarthy [11]). In this case, we can see on the surface that segments like to delete to whatever extent possible, even if full deletion will never be an option during the derivation.

In Hixkaryana, we will see parallels to McCarthy's examples in the sense that deletion does not seem to be an all-or-nothing response and that its variable extents appear on the surface. However, the nature of Gradual Deletion in Hixkaryana is apparently more complex, due to the way it manifests across different speech varieties. Specifically, we will consider the possibility that the grammars of certain varieties contain "crystallized" reduction steps from a Gradual Deletion chain that is synchronically active in only one variety. That "proto-variety" has clear motivations for certain reductions that are seemingly gratuitous in other varieties, namely prefix deletion and reduction to ʔ. We will analyze this crystallization process as arbitrary spillover of certain properties between varieties via inter-speaker borrowing through the course of language evolution.

The goal of this paper is thus twofold - for one, to exercise Gradual Deletion to its fullest extent in order to offer a formal hypothesis for previously unexplained patterns of null allomorphy and variable pre-nasal reductions in Hixkaryana; secondly, to advance the range of potential use cases for the Gradual Deletion framework by demonstrating how steps along the deletion gradient might bear independent weight in diachrony. At its most radical extent, the latter goal intends to feed theories in which synchronic and diachronic serial changes are not entirely separate. The shyness (gradualness) of a deletion pattern could accurately reflect the natural shyness of language change, in which something like debuccalization is the first step in the history of a given contextual sound loss. If such speculation accumulates more concrete evidence, it could produce a more cohesive picture of what deletion looks like in its broadest sense.

2 Nasality Agreement in Hixkaryana

2.1 Syncope, Denasalization, and Opacity

Hixkaryana shows a fairly pervasive rule of high vowel /i u/ syncope at morpheme boundaries (1), which apparently serves to improve brevity and alleviate hiatus.

(1) Syncope

a. /ni-ahe-no/	→ [n-ahe-no]	<i>he touched it</i>
b. /ki-rata-no/	→ [k-rata-no]	<i>I wept</i>
c. /ki-hananihi-jatʃkoni/	→ [k-hananih-jatʃkoni]	<i>he used to teach us</i>

Obstruent-Nasal sequences (henceforth KN) are generally unavailable in underlying forms in Hixkaryana. However, high vowel syncope has a tendency to generate KN, especially when the person/number-marking prefixes {ki-, ti-} or the past-tense marker /-no/ are involved. KN resulting from syncope is demonstrated by the partial derivations in (2).

(2) Syncope

a. /ki-niki-no/	→ k-nik-no	<i>I slept</i>
b. /ki-momoki-no/	→ k-momok-no	<i>I was waiting for you</i>
c. /ki-ɲake-jatʃhe/	→ k-ɲake-jatʃhe	<i>he will send us</i>
d. /ti-manho-no/	→ t-manho-no	<i>we danced</i>

What follows from these intermediate forms is often an unusual denasalization process whereby |KN| sequences transform to [KD]. D roughly maintains the place of its underlying N, so /n m ɲ/ become [d b dʲ] respectively. This process is the sole source of voiced obstruents in Hixkaryana. After denasalization, the rest of the person/number marker |k-| is removed (this only takes place before nasal-initial stems), giving us the outputs in (3). Boldface indicates denasalized segments.

(3) Denasalization (KD)

a. k-nik-no	→ [d ik -do]	<i>I slept</i>
b. k-momok-no	→ [b omok -do]	<i>I was waiting for you</i>
c. k-ɲake-jatʃhe	→ [dʲ ake -jatʃhe]	<i>he will send us</i>

This interaction is undeniably opaque, because [#D] (word-initial voiced stops) arise even in the surface-absence of the prefix obstruents responsible for triggering denasalization. D arises only due to the presence of the obstruent prefix, but persists after the prefix's removal. This overapplication may have been a useful cognitive adaptation that allowed listeners to infer the underlying presence of elided prefixes from the unique oral stops. Analytically, the opacity would provide a significant challenge for a mainstream full-parallel model. The only characteristic property of [#D] is its word-initial position, but there is no evidence that nasals should regularly denasalize word-initially in the language.

2.2 Variation

An alternative speech variety utilizes pre-nasal obstruent debuccalization to resolve *|KN| disagreement. This is reflected in (4), wherein no denasalization takes place and [ʔ] arises in the affected coda. This variant is peculiarly the only (non-ideophonic) context where glottal stop surfaces in Hixkaryana.

(4) Debuccalization (ʔN)

a. k-nik-no	→ [niʔ-no]	<i>I slept</i>
b. k-momok-no	→ [momoʔ-no]	<i>I was waiting for you</i>

There is also a variety in which both debuccalization and denasalization occur in tandem, visible in (5), as well as a variety (6) that shows no nasality agreement repair, in favor of preserving the fully-faithful form of the stem. In all variants however, we still see complete removal of prefixes (#K-N → #Ø-N).

(5) Double Repair (?D)

- | | | |
|----------------|--------------|------------------------------|
| a. k-nik-no | → [diʔ-do] | <i>I slept</i> |
| b. k-momok-no | → [bomoʔ-do] | <i>I was waiting for you</i> |

(6) Preservation

- | | | |
|--------------------|--------------|------------------------------|
| a. k-nik-no (KN) | → [nik-no] | <i>I slept</i> |
| b. k-momok-no | → [momok-no] | <i>I was waiting for you</i> |

2.3 Syncope-Blocking

The syncope that feeds these processes does not take place universally, which will prove critical for understanding why nasals are special in regard to reduction. Syncope is unconditionally blocked when the resulting sequence would be an obstruent cluster, lending us the unexpected pronounced prefixes in (7). This is strange because we know from our nasal-initial stems that prefixes are capable of deletion to resolve an awkward onset. On paper, there is no reason [Ø-K] should be an unacceptable output of /ki-K/ despite being a fine output of /ki-N/. This indicates sensitivity to the markedness of intermediate forms (or perhaps faithfulness related to intermediate processes), since surface forms do not differ in a way that would result in distinct blocking abilities.

(7)

- | | | |
|------------------|-----------------|---------------------|
| a. /ki-tajma-no/ | → [ki-tajma-no] | <i>I pushed you</i> |
| b. /ki-kita-no/ | → [ki-kita-no] | <i>I rushed</i> |

There may be look-ahead effects pertaining to highly-illegal tautosyllabic obstruent clusters in intermediate phases of the derivation. In other words, |#k-K| is so marked that it can never be output by /ki-K/ even if a theoretical subsequent repair is possible. By comparison, |K-N| is bad enough to cue the observed K-deletion, but also tolerable enough to occur intermediately, thus syncope is licensed for /ki-N/ forms.

A similar blocking effect occurs when syncope would result in an intermediate coronal cluster, either |t-n| or |t-ɲ|. As shown in (8), the nasal-related feeding effects can be obtained from the /ti-/ prefix only when it precedes non-coronal |b m| (8c). This makes enough sense not just within Hixkaryana (which never exhibits word-initial coronal clusters), but also across the Parukotoan family. In the closely related Wai Wai language, "tense consonants" (corresponding to the coronal and near-coronal section of the inventory) are never permitted to cluster (Hawkins [4]). Blocking by coronals is not a central observation here, but corroborates the hypothesis that hidden intermediate forms are affecting the potential for syncope to occur.

(8)

a. /ti-ɲake-jatʃhe/	→ [ti-ɲake-jatʃhe]	<i>I pushed you</i>
b. /ti-niki-no/	→ [ti-nik-do]	<i>I rushed</i>
c. /ti-manho-no/	→ [banho-no]	<i>we danced</i>

It is clear that nasals are unique when it comes to interaction with prefixes of the form /KV_{weak}/. While stem-initial nasals are sonorant enough to license syncope, they are not sonorant enough to surface with the resulting single-obstruent prefixes (*#K-N) - and crucially, this distinction must matter derivation-internally because there is no surface-true justification for the observed differences between stem-initial nasals and stem-initial obstruents.

The derivations in (9) summarize the asymmetries seen between stem types (obstruent, nasal, approximant⁺) and prefix types (coronal, non-coronal). Only the denasalizing variant is demonstrated here, but the relevant properties hold true regardless of which nasality agreement strategy is used by the speaker. In (9a) syncope is permitted and no prefix deletion is necessary because the resulting cluster is legal. In (9b) syncope is blocked to avoid an obstruent cluster. In (9c) syncope creates an illegal obstruent-nasal cluster, which first feeds denasalization, and secondly prefix-deletion. In (9d) syncope is blocked word-initially to avoid a coronal cluster, but still occurs word-medially and feeds denasalization, as in (9c). Note that blocking is always local, i.e. syncope may be permitted somewhere within a word but not elsewhere, which reinforces our assumption that the distribution of effects is systematic rather than suppletive.

(9)

a. /ki-rata-no/	→ k-rata-no	→ [kratano]	<i>I wept</i>
b. /ki-kita-no/	→ ki-kita-no	→ [kikitano]	<i>I rushed</i>
c. /ki-niki-no/	→ k-nik-no	→ [dikdo]	<i>I slept</i>
d. /ti-niki-no/	→ ti-nik-no	→ [tinikdo]	<i>we sleep</i>

3 Placelessness

3.1 Laryngeals in Hixkaryana

Recall that debuccalization as a nasality agreement repair (?N) is the only source of regular glottal stop in Hixkaryana. This is suspicious in and of itself, but especially so when we consider the broader context of laryngeal distributions in the language. [h] is extremely common as both an onset and coda, and occurs frequently before nasals at morpheme boundaries (10a,b) as well as morpheme-internally (10c,d). [hN] also occurs in derived environments, including from high vowel syncope (11).

(10)

a. w-ak-ɲoh-no	<i>I burned it</i>
b. wajwi-ɲanomkahnoh-no	<i>he rejected the arrows as no good</i>
c. ihme	<i>egg</i>
d. kahɲe	<i>one who creates</i>

(11)

- a. /o-horohi-ni-toko/ → [o-horoh-ni-toko] *when you stop*
b. /r-ompamnohi-ɲe/ → [r-ompamnoh-ɲe] *the one who teaches me*

This means the observed [ʔN] output stands blatantly contrary to the expected [hN], which seems like a more appropriate output given the facts of the language. It is not immediately clear why a glottal stop arises in this context. There cannot be any across-the-board rule mandating that pre-nasal laryngeals take the form [ʔ] because this would misgenerate (10). Further stipulating that the rule is unique to derived environments would persist to misgenerate (11). The data in (12) demonstrate an attested rule whereby /k/ reduces to [h] before other obstruents, which rules out the tempting possibility that [ʔ] results from the plosivity of the base ($s \rightarrow h$; $\kappa \rightarrow ʔ$), as opposed to the qualities of the environment.

(12)

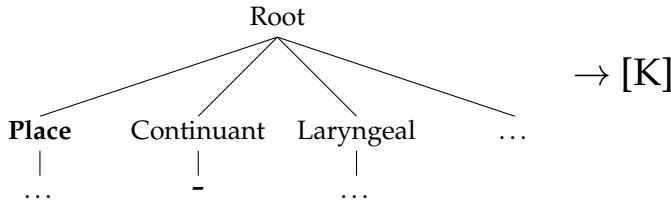
- a. /w-esnoku-su/ → |w-esnok-su| → [w-esnoh-su] *let me smell it*
b. /omoki-toho/ → |omok-toho| → [omoh-toho] *the time of his coming*

In combination, these facts paint the observed glottal stop as being somehow intrinsically linked to the nasality agreement system. It is only in this specific circumstance, wherein debuccalization is employed either as an alternative or supplement to denasalization, that [ʔ] surfaces, and it does so despite all external signs pointing toward realization as [h].

3.2 Feature Deletion

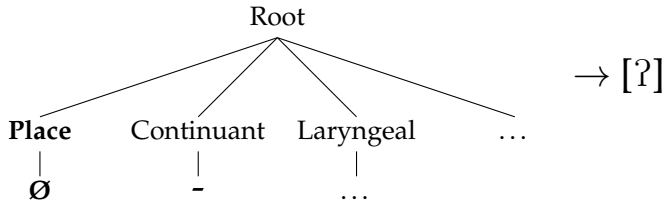
The source of this apparent laryngeal discrepancy is based in an assumption that because $\kappa \rightarrow [h]$ and $\kappa \rightarrow [ʔ]$ have distinct results, they must represent distinct reduction paths. This is the null hypothesis. An alternative hypothesis posits that the two distinct laryngeal outputs come from the same overarching linear reduction path, and that the different surface forms [ʔ h ∅] reflect minimally distinct extensions of that path. The basic presumed internal schema (modeled loosely after McCarthy [8]) for the at-issue segment is shown in (13). Specification for PLACE produces a supralaryngeal sound, giving us some sort of κ .

(13)



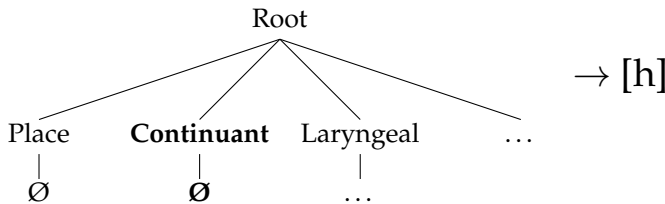
In (14), deletion of PLACE forces laryngeal activity to surface, and it must surface as [ʔ] rather than [h] due to the *minus* specification for CONTINUANT. This is a fairly uncontroversial example of debuccalization from Gradual Deletion as suggested by McCarthy.

(14)



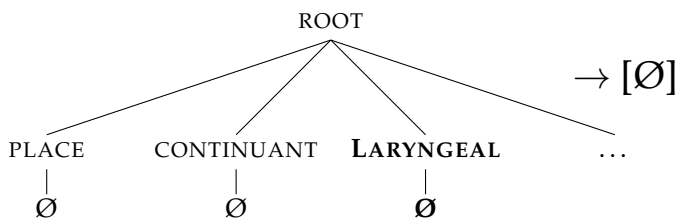
The following phase of deletion (15) may specifically target the segment's CONTINUANT node, which forces production of the default, unspecified laryngeal form. We have established that Hixkaryana laryngeals surface as [h] when all else is equal, so it is reasonable to assume that a geometry lacking specification for PLACE and CONTINUANT should result in [h].

(15)



In a full deletion case, the remainder of material (or perhaps the root node) is deleted from |h|, resulting in a null segment (16).

(16)



What this approach offers us is subtle but critical - different circumstances for reduction can motivate different extensions of a unified linear deletion process, meaning that we do not need to analyze debuccalization to [ʔ] as an idiosyncratic property of nasality disagreement sites. Rather, all laryngealization processes in Hixkaryana may contain intermediate |ʔ|, and nasality disagreement only differs in that it does not necessitate deletion of the CONTINUANT node.

It is reasonable to suspect that loss of PLACE suffices as a nasality pseudo-assimilation for some speakers because placeless segments are normally transparent to nasality (Walker [15]), and generally lack content that would otherwise impede coarticulation. This means debuccalization and denasalization are two sides of the same assimilatory coin and carry roughly equivalent advantages. The only difference is that for speakers of the [KD] variety, agreement is optimally achieved by forcing the nasal segment to agree with its predecessor, while for speakers of the [ʔN] variety, agreement is optimally achieved by revoking the pre-nasal segment's ability to disagree.

In derived laryngeal contexts besides nasality disagreement (12), $|\text{?}|$ is an insufficient repair. Further deletion can be motivated by some sort of persistent markedness at $|\text{?K}|$ sites that is not shared by $|\text{?N}|$. This could reasonably be a matter of stop clustering, as long as we ensure that such constraints would not also affect KD - perhaps correspondence is involved as a reduction facilitator, and $|\text{K-D}|$ do not correspond because of voice-disagreement. There are other potential ways to motivate $|\text{?}| \rightarrow [\text{h}]$, but regardless, accounting for the observed regular pre-obstruent reduction of $/\text{k}/$ would not be a significant challenge for Gradual Deletion.

3.3 Unifying Full and Partial Deletion

As mentioned previously, null allomorphs of person/number-marking prefixes occur before nasal-initial stems and, mystifyingly, only before nasal-initial stems. A gradual deletion framework may enable a working diachronic hypothesis of prefix deletion behaviors. If debuccalization can indeed be a direct reaction to nasality disagreement, then word-initial $|\text{#K-N}|$ can become $|\text{#?N}|$, which could easily collapse to $|\text{#?}|$ due to the natural weakness of a word-initial pre-consonantal glottal stop. The possibility of intermediate $|\text{#?N}|$ follows from the combined facts that denasalization overapplies word-initially when prefixes are missing (thus, intermediate $|\text{K-N}|$ must exist) and that pre-nasal debuccalization surfaces word-medially for many speakers (thus, $|\text{?}|$ is available).

Furthermore, we know that pronounced allomorphs exist for these prefixes, as in (7a) [ki-tajma-no], including before nasals when intermediate coronal clusters are involved, as in (8b) [ti-nik-do]. While these are technically different circumstances from $|\text{#K-N}|$, they are not surface-different as far as prefix deletion is concerned, because $[\text{K T N D}]$ are all grammatical word-initially. This corroborates the notion that the unavailability of $|\text{#KV-N}|$ was connected to the availability of $|\text{?N}|$ during some evolutionary stage in the language. $|\text{?N}|$ is responsible for facilitating a "clean deletion", and if a clean deletion is determined impossible, then the entire process (including syncope) is blocked. Conditional prefix deletion therefore resembles the model in (17), where the impossible intermediate forms in gray force the fully-faithful form to surface. In (17b), clean deletion is possible and thus a null prefix occurs.

(17)

a. /ki-kita-no/	→ k-kita-no	* k-kita-no	[ki-kitano]	<i>I rushed</i>
b. /ki-niki-no/	→ k-nik-no	→ ?ni?no	→ [ni?no]	<i>I slept</i>
c. /ti-niki-no/	→ t-nik-no	* t-nik-no	[ti-nikdo]	<i>we sleep</i>

Under this hypothesis, the unique properties of Hixkaryana nasals as they pertain to prefix deletion and debuccalization are treated as minimally different consequences of active conditions during a linear series of featural deletions of the shape $\text{K} \rightarrow \text{?} \rightarrow \text{h} \rightarrow \emptyset$. The specific surface form depends on precisely when convergence takes place during the series. This alleviates the need for separate treatments of pre-nasal prefix deletion, pre-nasal glottalization to $|\text{?}|$, and pre-obstruent glottalization to $[\text{h}]$. Importantly, the series also enables a more complete explanation for the unique emergence of $|\text{?}|$ before affix nasals.

Stem-internal $|\text{?N}|$ is not surface-available for all speakers, so intermediate debuccalization is most likely not synchronically responsible for all instances of the null-prefix process. More generally, there is no satisfying synchronic explanation. A traditional, segment-based deletion

theory might say that prefixes delete at the $|K-N|$ stage simply to resolve the complex onset, but it would then not be clear why the deletion option is unavailable before obstruent-initial stems (17a). The uniqueness of nasals with respect to null prefixes indicates that the null forms are an evolutionary artifact. Speculatively, the debuccalizing variant $[?N]$ could be left over from a time in the language's history when pre-nasal obstruent glottalization was the sole mechanism for nasality agreement, or it could have resulted from a mass appropriation of characteristics of the debuccalizing variant by other speakers.

The cross-speaker uniformity of prefix deletion is likely also related to the general weakness of inflectional morphology. The prefixes at issue do not contribute to the core meanings of the words they attach to, so they are easier to elide for the sake of brevity in day-to-day language use (McCarthy & Prince, 1995). This principle gives some grounding to the appropriation possibility, as the rest of the Hixkaryana population conceivably could have imitated the null prefix produced by $[?N]$ -variety speakers.

3.4 Interim Summary

The Hixkaryana data have shown us two main processes of interest - pre-nasal glottalization to $?$ (variable) and pre-nasal deletion of person/number-marking prefixes (universal). The core hypothesis here has been that the treatment of these processes should be unified in the grammar, with the goal of unification being threefold: 1) to allow a more restrictive grammatical architecture that does not need to make reference to separate, similar processes; 2) to satisfy reasonable expectations about the relationship between segment reduction (partial deletion) and full deletion; and 3) to advance a concise speculation for the evolutionary emergence of universal prefix deletion in the language.

In accordance with the third goal, the working diachronic hypothesis is based on the observation that prefix deletion is universal yet lacks a concrete motivation. Because of the combined facts that prefix deletion shares a unique environment with glottalization and that glottals are notoriously unstable (are known to be the penultimate step in a gradual deletion chain), it seems reasonable to suggest that the prefix deletion seen here is an artifact of pre-nasal glottalization. Therefore, out of the four variants documented $\{KN KD ?N ?D\}$, the $[?N]$ version is best treated as the specific proto-variety from which the others split, because it contains surface pre-nasal glottal stop and therefore has a concrete motivation for why prefixes would be unstable pre-nasally.

In terms of representation in the synchronic grammar, it would be cumbersome and probably dishonest to attempt to derive non-glottalizing variants using only constraints that encode phonetic and objective cognitive biases. Under the hypothesis that we are seeing an evolutionary artifact that lacks synchronic grounding in the broader context of the language, then there should be constraints encoding an arbitrary preference. In this case, there should be a way to enforce conformity to the underlying structure of a variant that a given speaker does not use themselves. While all motivations must ultimately make sense and be present in a non-arbitrary fashion in some varieties, certain constraints in other varieties should be permitted to retrieve their motivation indirectly on the basis of cross-reference to the proto-variety. This is essentially similar to what we see in established correspondence mechanisms such as Output-Output Faithfulness (Van Oostendorp [14]) and Lexical Conservatism (Steriade [13]), with the difference being that in this case, word forms may correspond across speech varieties. This will allow a more honest and direct representation of the working diachronic hypothesis in which we see variants seemingly defying their local grammars due to a lingering property inherited from an ancestor (or cohabitant).

4 Implementation

4.1 Tools

As proof that Gradual Deletion is viable for capturing the whole range of attested variants, each variant should be derived within the framework, using conceptually honest constraints, and each should differ from one another as little as possible so as to be realistic close evolutionary sisters. In Optimality-Theoretic (OT) terms, we should presume that constraint rankings differ between varieties only when there is empirical evidence for a distinction.

Gradual Deletion is situated within Harmonic Serialism (HS) (McCarthy [10]), a framework in which a monotonic (one change per output) series of OT-regulated changes occurs until the most optimal form is output. For example, gradually deleting κ entails two input-output relationships, one from κ to ? and one from ? to $\emptyset$. The output of each tableau provides the input of the following one, and derivations conclude when the fully-faithful candidate is selected (i.e. there are no optimizing changes left to be made), which is called "Harmonic Convergence". Because the present data will rely on representing multiple processes in a single derivation, their respective HS derivations may be fairly long (up to 6 steps).

An HS system is further complicated by "Hidden Structure" problems wherein multiple distinct serial paths (consisting of distinct intermediate outputs) can ultimately result in the same surface form. Hidden Structure problems are of special concern to people studying phonological learning (Jarosz [6]), but in our case will only mean that representations are not quite as straightforward. As we will see in sections 4.5-4.7, some inputs will produce multiple winning candidates that do not actually differ in viability. In these cases, one of the winners will be selected arbitrarily for conciseness.

4.2 Gratuitous Reduction and Output-Output Faithfulness

A pressing challenge for the analysis is the treatment of apparently gratuitous reduction to ? . This occurs in the $[\text{?D}]$ case, wherein there is a peculiar overlap of nasality agreement solutions, but also takes place in prefix deletion in all but the $[\text{?N}]$ case. To briefly recap, the working hypothesis of prefix deletion so far has been that prefixes delete pre-nasally due to an attempted glottalization in the same fashion as stem-internal pre-nasal glottalization, which presumably mitigates the markedness of $|\kappa N|$. That is, prefix deletion takes the shape $/\#\kappa N/ \rightarrow |\#\text{?}N| \rightarrow [\#\emptyset N]$. An intermediate glottal form for prefixes explains why syncope is not blocked before nasals despite that the resulting sequence is illegal - the glottal provides an escape route that is not available before obstruents, which is why $|\#\kappa\emptyset\kappa|$ is blocked. The glottal naturally collapses due to its weak position, giving us the observed null inflection.

The problem with this analysis is that it can only apply synchronically to the variety in which stem-internal debuccalization is a legal nasality agreement mechanism and denasalization is not, i.e. $[\text{?N}]$. In this variety, prefix collapse is a natural consequence of constraints needed to derive the stem (see 4.5). In other varieties, attempting to derive prefix collapse according to the process described above would incidentally result in unwanted debuccalization in the stem.

This situation almost demands a stratal approach, in which there would be separate grammars for simplex and complex forms, with only the complex forms would having the arrangements to motivate reduction as a nasality agreement mechanism. In principle, this would be viable for the $[\text{KD}]$ variety because nasality agreement in the root would have already taken place

(via denasalization) by the time affixes are added. However, a contradiction would remain for the faithful KN variety in which disagreement is permitted in the root. For this variety, any motivation to reduce at the complex level would also affect the root contained within. In other words, debuccalization would be counterbled by the absence of a denasalizing option.

Another way to separate roots from affixes is with an Indexed Constraint Theory (ICT) (Inkelas [5]) or Positional Faithfulness (Beckman [1]) framework, either of which permit constraints to be directly indexed to certain morphological components. Indexing constraints would solve the problem whereby constraints on complex forms affect roots in unwanted ways, because constraints would instead be tied exclusively to affixes. However, this could quickly become clunky when we consider that multiple processes would need to be split to enable the full range of variants. Nasality agreement (M) must be split between root and affix levels in order to elicit uniform prefix reduction, but so must nasal identity (F) in order to prevent denasalization from bleeding debuccalization at the prefix level (if this was not enforced, we would see [#KD] surface so long as denasalization is a valid solution). This version of the theory also does not get us any closer to understanding the [ʔD] variety, in which there is gratuitous reduction within the stem.

If pre-nasal prefix deletion is best analyzed as an evolutionary artifact left behind by the very same strain of debuccalization that affects the [ʔN] variant, then a more direct and cohesive way of expressing such a relationship is by forcing cross-variant loyalty, i.e. all variants should follow suit with respect to the presumed proto-variant [ʔN]. This relationship could be represented using Output-Output (OO) Faithfulness (Van Oostendorp [14]) between varieties, permitting us an OO constraint for place deletion such that if place is removed from segment α_1 in the inflection of the [ʔN] variant, then place should also be removed from corresponding segment α_2 in other variants. Specification for the affix level could be stipulated in the fashion of the aforementioned ICT system (FAITH-OO_{AFF}, 4.3), giving us a single constraint able to deal with the prefix deletion problem on its own.

The conspicuous drawback of this approach is that such a constraint is convoluted and requires borrowing from multiple theories. However, it does not do so carelessly. If the artifactual hypothesis has any weight, then we are essentially admitting that something in the non-[ʔN] varieties is explicitly out-of-line with the rest of the speaker's synchronic grammar. Enforcing mimicry of the proto-variant is a very direct way of expressing that a property has been "left behind", lingering arbitrarily in other variants.

Furthermore, an OO constraint split between the root and affix level uniquely allows generation of the [ʔD] case via the ROOT version of the constraint. If the [ʔD] variety is mostly identical to [KD] except with an added pressure to debuccalize root-internally in accordance with [ʔN], it will naturally manifest as [ʔD]. This makes for a solution that is very cohesive in its treatment of gratuitous reduction cases.

The suggestion that affixes are uniformly sensitive to correspondence despite that roots only show this property in the [KD] variety falls in line with the Root-Affix Faithfulness Metaconstraint (McCarthy & Prince [7]), which posits that IO (input-output) faithfulness is more important in roots than in affixes. This is especially evident in our case, wherein inflectional morphology is lost entirely and uniformly across varieties, demonstrating intrinsic weakness. Because OO Faithfulness is pressuring IO violation, it makes sense for the former to be more active precisely when the latter is less active.

In summary, the gratuitous reduction cases can be jointly conceptualized as (synchronically) arbitrary mimicry of the proto-variant [ʔN], represented using OO Faithfulness. At the affix level, OO Faith is universal, giving us ubiquitous prefix deletion. At the root level, OO Faith only holds

in the [ʔD] variant, giving us redundant glottalization after denasalization has occurred.

4.3 Constraints

Let nasality agreement be driven by the constraint (18) $\text{AGREE}[\text{NAS}]_{\text{GRAD}}$, formulated in a gradual fashion after (McCarthy [11])’s approach to vowel deletion in which the precise badness of a single marked segment is represented as a quantity of violations. While McCarthy leveraged reference to the [COLOR] and OPEN features borne by a vowel, we will reference the ROOT and PLACE features borne by a consonant. This maneuver effectively dictates that when a consonant appears in a weak (nasality-disagreeing) context, it is better for it to be placeless (glottal) than placed, but it will still be marked unless it is removed entirely.

This is important for the purpose of avoiding premature reduction of prefixes. If denasalization and debuccalization were equally powerful repairs for nasality disagreement, and prefixes are independently motivated to reduce, then a debuccalizing solution will always outrank a denasalizing solution in prefixes for its ability to feed two birds with one worm. The only straightforward way to avoid this, it seems, would be to let the markedness of unreduced prefixes be outranked by a faithfulness check on place deletion (reduction), which would effectively render prefix deletion impossible by the nature of the $F \gg M$ configuration.

It is also conceptually reasonable for debuccalization to be treated as an imperfect repair. As noted in 3.2, pre-nasal laryngeals are preferable to supralaryngeals because of their transparency. The [ʔN] solution successfully escapes [-NAS][+NAS], but no actual assimilation takes place. By contrast, denasalization (from which KD results) shows [-NAS][-NAS] agreement by way of assimilation, and thus harmony is achieved without a need for further phonetic work in the form of nasality onset determination.

(18)

$\text{AGREE}[\text{NAS}]_{\text{GRAD}}$ Assign one violation for every consonantal ROOT or PLACE node in segment α such that $[\alpha \text{ NAS}][-\alpha \text{ NAS}]$.

Because some variants do not denasalize, a boilerplate faithfulness constraint regulating nasality loss (19) is also necessary.

(19)

$\text{IDENT}[\text{NAS}]$ Assign one violation for every change in $[\pm \text{NAS}]$.

As discussed in 4.2, correspondence will be leveraged to force reductions that are not motivated locally. The [ʔN] variety influences all others at the affix level, and influences the [ʔD] variety at the root-internal level. These effects can be mediated by constraints (20, 21) $\text{DEP}(\text{PLACE})\text{-OO}$, which enforce that PLACE deletes to whatever extent occurs in the [ʔN] variant. The intended implication of the use of DEP here is that if a variant neglected to place-delete when appropriate, it would functionally be inserting place relative to its place-lacking correspondent. The two versions of $\text{DEP}(\text{PLACE})\text{-OO}$ are indexed to the root and affix portions of participating inputs directly, as is done in (McCarthy [9]).

(20)

DEP(PLACE)-OO_{RT} Assign one violation for every root consonant in α with additional content relative to its mirror segment in correspondent β (the [?N] variant).

(21)

DEP(PLACE)-OO_{AF} Assign one violation for every affix consonant in α with additional content relative to its mirror segment in correspondent β .

There must also be a constraint motivating removal of prefixes that have already been reduced. Again following McCarthy [9], we will employ (22) HAVEPLACE, which penalizes placelessness, thereby encouraging glottals to collapse into null segments. This process is naturally predicted by the principle that placeless segments are weak, which informs the core of the Gradual Deletion framework.

(22)

HAVEPLACE Assign one violation for every segment lacking a PLACE node.

On the flip side, we do not want all glottals to collapse, particularly root-internally. The fact that we see full deletion in prefixes (onset position in this case) and not in roots (coda position) shows that there are active pressures against removal of some segments that are integral to word structure. This could potentially have to do with mora retention / syllable structure, but for the sake of not involving prosody we will treat it as a similarly plausible matter of contiguity faithfulness (McCarthy & Prince [7]) (23), i.e. segments that are not in contact in the input should not make contact in the output either. Deleting segments word-initially does not pose a problem for contiguity, but deleting in between segments is marked.

(23)

CONTIGUITY Assign one violation for every pair of touching segmental indices that did not touch in input.

We will test our implementation on the four variants of the underlying form /ki-niki-no/ (Sec 2), because this word contains two sites where nasality agreement will become an issue following syncope, stem-internal and external.

For conciseness, derivations will begin at the post-syncope stage |k-nik-no| because syncope is invariant and not of utmost interest. The only mechanism that would be necessary to ensure syncope takes place before other processes is an indomitable constraint of the form *V_{weak}, to penalize the existence of weak vowels and force their removal before anything else takes place. This constraint is implicitly responsible for providing the input |k-nik-no|, although the true underlying form is indeed /ki-niki-no/.

There are also some constraints that will be left implicit due to inactivity, namely faithfulness constraints that do not exhibit any power in our situation - MAX(PLACE) (which would theoretically regulate place deletion), MAX(ROOT) (which would regulate full deletion of unplaced segments), and MAX(V) (which would regulate syncope).

4.4 Denasalizing Variant

In the [KD] variant, there is a strict linear ranking hierarchy between AGREE[NAS], DEP(PLACE)-OO, and HAVEPLACE. There must be dominance of DEP(PLACE)-OO » HAVEPLACE in order to permit place deletion. If the reverse was true, all place deletion motivated by DEP(PLACE)-OO would be blocked.

There also must be dominance of AGREE[NAS] over DEP(PLACE)-OO in order to prevent premature place deletion, which would bleed denasalization. AGREE[NAS]’s gradual formulation (18) makes it such that denasalization is the more powerful mechanism for achieving nasality agreement, however it is theoretically no more powerful than full deletion of a pre-nasal segment. Therefore, if word-initial [ʔ-N] is allowed to exist, no constraint ranking would be able to prevent collapse into [Ø-N], which would make any attempt at denasalization to [Ø-D] a gratuitous violation of ID(NAS). Ensuring that denasalization precedes prefix deletion therefore requires that AGREE[NAS] » DEP(PLACE)-OO.

(24) Derivation of *dikdo*.

a. First denasalization (order irrelevant).

/k-nik-no/	AGREE[NAS] _{GRAD}	DEP(PLACE)-OO _{AF}	HAVEPLACE
a. k-nik-no	***!*	*	
☞ b. k-dik-no	**	*	
☞ c. k-nik-do	**	*	
d. ʔ-nik-no	***!		*

b. Second denasalization.

/k-dik-no/	AGREE[NAS] _{GRAD}	DEP(PLACE)-OO _{AF}	HAVEPLACE
a. k-dik-no	*!*	*	
☞ b. k-dik-do		*	
c. ʔ-dik-no	*!*		*
d. k-diʔ-no	*!	*	*

c. Reduction of prefix, motivated by DEP(PLACE)-OO.

/k-dik-do/	AGREE[NAS] _{GRAD}	DEP(PLACE)-OO _{AF}	HAVEPLACE
a. k-dik-do		*!	
☞ b. ʔ-dik-do			*
c. k-diʔ-do		*!	*

d. Full removal of prefix, motivated by HAVEPLACE.

/ʔ-dik-do/	AGREE[NAS] _{GRAD}	DEP(PLACE)-OO _{AF}	HAVEPLACE
a. ʔ-dik-do			*!
☞ b. Ø-dik-do			

e. Harmonic convergence.

/dik-do/	AGREE[NAS] _{GRAD}	DEP(PLACE)-OO _{AF}	HAVEPLACE
☞ a. dik-do			
b. diʔ-do			*!

4.5 Reducing Variant

In the [ʔN] variant, prefix deletion is motivated intrinsically because nasality faithfulness and nasality agreement are both prioritized. Therefore, the DEP(PLACE)-OO constraints are not active because correspondence would be self-referential and redundant.

AGREE[NAS] » HAVEPLACE is a necessary ranking to permit reduction as an agreement repair, and ID(NAS) » AGREE[NAS] is necessary to prevent denasalization, the more powerful agreement mechanism, from occurring instead. CONTIGUITY is similarly undominated, which allows harmonic convergence to occur upon [niʔ-no] despite that it does not show perfect agreement according to our formulation of AGREE[NAS].

(25) Derivation of *niʔno*.

a. Reduction of prefix or root-internal K (order irrelevant).

/k-nik-no/	CONTIGUITY	ID(NAS)	AGR[NAS] _{GRAD}	HAVEPL
a. k-nik-no			****!	
☞ b. ʔ-nik-no			***	*
☞ c. k-niʔ-no			***	*
d. k-dik-no		*!	**	

b. Deletion of prefix, motivated by HAVEPLACE.

/ʔ-nik-no/	CONTIGUITY	ID(NAS)	AGR[NAS] _{GRAD}	HAVEPL
a. ʔ-nik-no			***!	*
☞ b. Ø-nik-no			**	
c. ʔ-niʔ-no			**	*!*
d. ʔ-nik-do		*!	*	*

Note that this arrangement is not fully honest to the derivation's Hidden Structure, because candidate C would lead to the same eventual outcome as candidate B. Its status as loser here is purely superficial.

c. Reduction of root-internal K for further agreement.

/nik-no/	CONTIGUITY	ID(NAS)	AGR[NAS] _{GRAD}	HAVEPL
a. nik-no			**!	
☞ b. niʔ-no			*	*
c. nik-do		*!		

d. Harmonic Convergence. Violations in AGREE[NAS] are allowed because of CONTIGUITY.

/niʔ-no/	CONTIGUITY	ID(NAS)	AGR[NAS] _{GRAD}	HAVEPL
☞ a. niʔ-no			*	*
b. niØ-no	*!			
c. niʔ-do		*!		*

4.6 De-nasalizing+Reducing Variant

Because we are positing that the [ʔD] variant gets its glottal stop from correspondence, its structure should be similar to that of the [KD] variant, with the addition of DEP(PLACE)-OO_{RT}. For the same reasons as DEP(PLACE)-OO_{AF}, the ranking of DEP(PLACE)-OO_{RT} must be above

HAVEPLACE (to permit reduction) and below AGREE[NAS] (to avoid bleeding denasalization). See 4.5 for the precise argumentation.

Much like its [ʔN] cousin, this variant also depends on CONTIGUITY in order to prevent root-internal collapse of $|\text{ʔ}| \rightarrow [\emptyset]$, which would otherwise occur for the sake of HAVEPLACE.

(26) Derivation of *diʔdo*.

a. First denasalization (order irrelevant).

/k-nik-no/	CONTIG.	AGR[NAS] _{GRAD}	DEP(PL)-OO _{RT}	DEP(PL)-OO _{AFF}	HAVEPL
a. k-nik-no		***!*	*	*	
☞ b. k-dik-no		**	*	*	
☞ c. k-nik-do		**	*	*	
d. ʔ-nik-no		***!	*		*
e. k-niʔ-no		***!		*	*

b. Second denasalization.

/k-dik-no/	CONTIG.	AGR[NAS] _{GRAD}	DEP(PL)-OO _{RT}	DEP(PL)-OO _{AFF}	HAVEPL
a. k-dik-no		*!*	*	*	
☞ b. k-dik-do			*	*	
c. ʔ-dik-no		*!*	*		*
d. k-diʔ-no		*!		*	*

c. Reduction of root or prefix (order irrelevant).

/k-dik-do/	CONTIG.	AGR[NAS] _{GRAD}	DEP(PL)-OO _{RT}	DEP(PL)-OO _{AFF}	HAVEPL
a. k-dik-do			*!	*	
☞ b. k-diʔ-do				*	*
☞ c. ʔ-dik-do			*		*

d. Reduction of prefix (or root, if prefix was reduced first).

/k-diʔ-do/	CONTIG.	AGR[NAS] _{GRAD}	DEP(PL)-OO _{RT}	DEP(PL)-OO _{AFF}	HAVEPL
a. k-diʔ-do				*!	*
☞ b. ʔ-diʔ-do					**
c. k-diØ-do	*!			*	

e. Removal of prefix, improving satisfaction of HAVEPLACE.

/ʔ-diʔ-do/	CONTIG.	AGR[NAS] _{GRAD}	DEP(PL)-OO _{RT}	DEP(PL)-OO _{AFF}	HAVEPL
a. ʔ-diʔ-do					**!
☞ b. Ø-diʔ-do					*
c. ʔ-diØ-do	*!				*

f. Harmonic convergence.

/diʔ-do/	CONTIG.	AGR[NAS] _{GRAD}	DEP(PL)-OO _{RT}	DEP(PL)-OO _{AFF}	HAVEPL
☞ a. diʔ-do					*
b. diØ-do	*!				

4.7 Faithful Variant

In the [KN] variety, AGREE[NAS] is not an active constraint. The only relevant change made to the UR is prefix deletion, the mechanism for which is the same as in other variants. For conciseness, denasalizing candidates are omitted because they lack any motivation.

(27) Derivation of *nikno*.

a.

/k-nik-no/	DEP(PLACE)-OO _{AFF}	HAVEPL
a. k-nik-no	*!	
☞ b. ʔ-nik-no		*
c. k-niʔ-no	*!	*

b.

/ʔ-nik-no/	DEP(PLACE)-OO _{AFF}	HAVEPL
a. ʔ-nik-no		*!
☞ b. Ø-nik-no		
c. ʔ-niʔ-no		*!*

c. Harmonic convergence.

/nik-no/	DEP(PLACE)-OO _{AFF}	HAVEPL
☞ a. nik-no		
b. niʔ-no		*!

5 Discussion

5.1 Alternatives

The approach advanced here involves a fairly radical melange of tools with high generative power (Harmonic Serialism, Gradual Constraints, Output-Output Faithfulness), which makes it especially deserving of scrutiny. The simplest, least powerful alternatives should be discussed first.

A pure-parallel model ("vanilla" OT) was dismissed earlier on the grounds that it would struggle to account for the opaque overapplication of denasalization. There is no parallel motivation for /KV-N/ to become [Ø-D], unless nasal segments were universally pressured to become oral word-initially, which is not the case. Stratal OT, a more mainstream framework, would fare better at handling this problem, but only because of its serial component. There do not seem to be any potential gains from a mid-derivational constraint re-ranking in this case, so Harmonic Serialism is a more honest (and arguably more computationally efficient) solution.

A traditional SPE-style linear rule-ordering framework could address the opacity problem, but might struggle with the blocking effects discussed in 2.3. In particular, the syncope rule would have to stipulate that weak vowels do not delete when preceding obstruents, and do not delete in between coronals. Having to represent the "elsewhere case" makes for a convoluted system (28) which depends on defining multiple, similar environments (or possibly having two similar rules) in which surrounding segments must either match negative for [COR] or mismatch for [COR], and must precede a sonorant.

(28)

/i u/ → [Ø] / [-COR] _ [-COR, SON] ; [α COR] _ [-α COR, SON]

Hypothetically, this would be followed by an optional denasalization rule followed by an obligatory cluster reduction rule to remove the remainder of the prefix. While this solution would certainly beat the present analysis in terms of overall analytical brevity, it lacks any implementational richness. It can be deduced from the rule in combination with external facts of Hixkaryana that only nasal-initial stems would result in null prefixes, but it does not help us understand why such an exception would arise.

Presuming instead a Harmonic-Serial approach, there are alternatives to discuss in respect to the specific handling of prefix deletion. When faced with an intermediate form |K-N|, the most obvious explanation for the disappearance of K is avoidance of complex onsets. However, the fact that syncope is blocked before obstruent-initial stems indicates that deletion via *COMPLEXONSET is not available. On paper, if /K-N/ → [Ø-N] is possible, then so should be /K-K/ → [Ø-K]. This line of reasoning was used in 3.3 to tie prefix deletion to nasality agreement, with the hypothesis being that prefix deletion is in fact an accidental consequence of agreement in the proto-variety [?N]. We also see that avoidance of complex onsets is not a priority in sentences like (29), which demonstrates the legality of word-initial obstruent clusters in non-derived environments.

(29)

[tkatʃhonke rmahafa naha Waraha]

Waraha has lots of things. (Derbyshire [3])

This does not necessarily mean that *COMPLEXONSET plays no role at all. Indeed, one possible alternative to a prefix deletion implementation based on Output-Output faithfulness is to invoke a "ganging up" effect by way of combining *COMPLEXONSET and AGREE[NAS] in a Harmonic Grammar (Legendre et al., 1990) framework. The unique environment that deletes prefixes is exactly that in which there is a complex onset composed of an oral and nasal segment. We cannot permit *COMPLEXONSET to be ubiquitous, but we could arrange for it to become active when it intersects with nasality disagreement, which would yield perfect prefix deletion.

The downsides of such an approach are that, one, Harmonic Grammar is very powerful and arguably overkill for this problem, and two, it could not independently eliminate the need for Output-Output faithfulness, because of the OO system's additional utility in deriving coda reduction in the [?D] variety. Part of the architectural reasoning behind using an OO constraint for prefix deletion was that it simultaneously explains the root-internal and inflectional cases of gratuitous reduction. Refusing this opportunity in favor of Harmonic Grammar would unnecessarily convolute the analysis.

Deriving [?D] without the opportunity to cross-reference other varieties using OO would be difficult, but not impossible. The established problem is that pre-nasal debuccalization is completely redundant when intersecting with denasalization, because [KD] fully satisfies nasality agreement. However, it is possible that this redundancy results from some sort of imperfectly-optimizing serial process, whereby the more modest change |?N| occurs first, then the grammar "re-evaluates", detects that nasality agreement is still not fully satisfied, and goes on to make the bolder change [?D]. The problem with this hypothesis is that such an operation could not be effectively grounded in the synchronic grammar. On the basis of three constraints that respectively enforce nasality agreement (M), regulate place deletion (F), and regulate denasalization (F), no

ranking can permit denasalization to occur while simultaneously preventing $|KD|$ from bleeding place deletion. The only escape from this conundrum would be overhauling the gradual calculation of nasality agreement, which would create many more problems than it would solve.

Another possibility is that $?$ appears as an optional consequence of denasalization, and in fact has no connection to nasality agreement at all. The fact that surface glottal stop is unique to this process might tempt us to believe that pre-glottalization is a characteristic property of voiced stops (which exist exclusively as allophones of nasals), and it is conceivable that $|KD| \rightarrow [?D]$ is essentially the "not-voiceless" equivalent of Hixkaryana's $|kT| \rightarrow [hT]$ process. There is no voiced laryngeal in Hixkaryana, so if we assume that $?$ is voice-unspecified, it could be superior to $[h]$ when preceding D while also improving articulation relative to $|KD|$.

However, this hypothesis is intuitively far-fetched in the context of the 2x2 system of attested variants, where we see that every combination of $\{\pm \text{denasalization}, \pm \text{glottalization}\}$ occurs. The claim would then have to be that glottalization in the $[?N]$ variant is motivated independently of that in the $[?D]$ case. Recall that pre-nasal glottal stop is an unexpected guest, with the standard choice for pre-nasal laryngeals in Hixkaryana being $[h]$. Therefore, the prior reasoning for why $[?]$ might appear before D is non-extensible to $[?N]$. Knowing that $[?N]$ is most likely related to nasality agreement, it does not seem realistic nor efficient to posit that the closely related variant $[?D]$ has different motivations for reaching essentially the same outcome, which puts the diachronic hypothesis and its OO representation in a better position.

5.2 Implications

This analysis was intended as an exercise in using a gradual understanding of deletion to explain processes that are otherwise mysterious and whose motivations are unclear. We first saw that two distinct processes occurring before nasals in Hixkaryana, denasalization and debuccalization, are best analyzed as jointly motivated by nasality agreement. That premise was combined with an established principle of gradual segmental weakness to speculate that prefix deletion is an extension of agreement-motivated pre-nasal debuccalization, fulfilling our expectations of hierarchical weakness while providing a possible explanation for the null allomorph, whose presence was previously treated as idiosyncratic.

That resulting hypothesis sought to highlight the utility of gradual deletion as a mechanism for explaining missing segments. Environments where mysterious ellipsis occurs should be cross-referenced with environments where glottalization occurs, in case there is likelihood of the null element being a "collapsed" laryngeal. This type of investigation has potential utility in a broad range of problems related to determining the source of ellipsis.

More specifically, future research should be open to the possibility that reduction for strict assimilatory purposes may feed deletion related to weakness. In our case, reduction presumably serves to remove an oral specification that conflicts with a nasal one, or perhaps to decrease the volume of disagreeing content (the exact analysis therein is unimportant). K is not naturally weak, but it can become weak in order to agree with N , feeding its own collapse. This line of logic can be applied to myriad situations that result in glottalization, giving insight into why deletion is sometimes chosen over identity changes when adjacent segments disagree.

The implementation of the working hypothesis intended to demonstrate the useful connection between gradual deletion, language evolution, and variation. We saw that all variants showed prefix deletion (presumably an extension of debuccalization), but only some variants showed root-internal debuccalization, and none showed complete deletion within roots.

This data could give us a lot of information about the diachrony of pre-nasal reductions in Hixkaryana. The ubiquity of prefix deletion was used to speculatively diagnose [ʔN] as a proto-variant, on the basis that there is only motivation for such prefix deletion to occur so long as there is a pressure to debuccalize pre-nasally. This is to be contrasted with root-internal debuccalization, which remains in the [ʔN] and [ʔD] variants, despite that the latter seems to lack any synchronic motivation for it. Under the presumption of correspondence with respect to [ʔN] (split between a root and affix version), there is a sort of competition between OO and IO faithfulness whereby OO faith is motivating reduction of segmental content in accordance with [ʔN] while IO faith motivates retention of content. OO Faith is dominant in affixes in part due to the activity of the Root-Affix Faithfulness Metaconstraint, which helps us understand why the null allomorphy has persisted through divergent varieties. The regular, derivationally "late" nature of inflectional morphology would lead us to believe that inflection should apply and be pronounced whenever applicable, but an exception arises here as a possible extension of OO faith's competitive status. Because OO faith is less dominant in roots, we see liberal divergence from [ʔN]'s agreement strategy, including both negligence to agree (as in [KN]) as well as the alternative denasalizing strategy [KD]. Debuccalization is sticky in [ʔD], which indicates that root-level OO faith is still at least somewhat active. Speakers sometimes conform to the root structure of the proto-variant even when it (arguably) serves no articulatory purpose.

These cases of gratuitous reduction, namely prefix deletion and [ʔD], can be conceptualized as crystals of gradual deletion, a dynamic process rendered static by evolution. This lens intends to build upon the primary takeaway that gradual deletion offers insight into apparent phonological exceptions related to null elements. Specifically, a $\emptyset$ that seems idiosyncratic in the local phonology of its environment might have inherited reduction from a related form, and subsequently collapsed due to its artificially weak state.

5.3 Imprecision Disclosure

Data for Hixkaryana are limited, and even more so for the specific phenomena discussed here. The language has not received direct documentation since Derbyshire's work 40-ish years ago, the focus of which was (morpho)syntax rather than phonology, and there does not seem to be any recorded sound data available for the language. It is also unspecified in Derbyshire's work whether the documented variation occurs between or across speakers, or both. It is likely that the patterns at issue are more nuanced and perhaps noisier than what Derbyshire was able to express in a paragraph.

Because both glottalization and (de)nasalization can be partial, and because we are dealing with coarticulatorily motivated patterns, it is particularly likely that the level of the segment is broader than the true scale of these operations. In the de-nasalizing variety, orality might be spilling backward from the suffix onto only the final subsegment, which intuitively seems more likely than a full-segment loss of nasality, due to the typological rarity of denasalization. If this is indeed the case, then a subsegmental framework like Q Theory (Inkelas & Shih [12]) would come in handy for building a clearer phonetic picture.

Similarly, speakers of the debuccalizing variety may only be debuccalizing partially, which would fall more in tune with Hixkaryana's lack of other ʔ contexts. If both of these possibilities are accurate, the resulting image is more of a spectral one - speakers make modest accommodations to unwanted |KN| sequences, and speakers differ in exactly how pronounced each accommodation will be. It is not clear from Derbyshire's report whether / to what extent this interpretation is

realistic, as opposed to a discrete set of well-defined, consistent varieties.

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